Laboratory 12

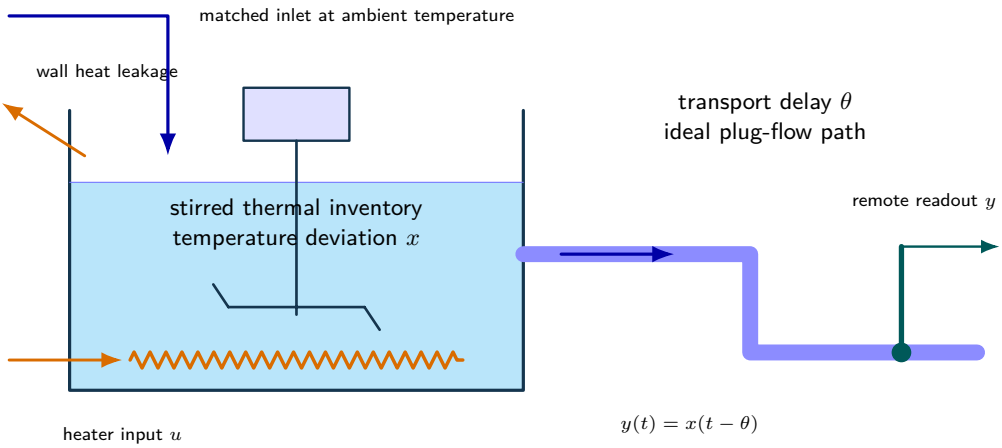
A synthetic process from identification to validation

Worked solutions – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Synthetic process: $G(s) = 1.8e^{-0.6s}/(4s+1)$. Identification and validation use separate generated records. No parameter estimate or fitted response is a physical measurement.

1. Inferring an aggregate thermal process from a step

The illustration shows a well-mixed thermal inventory followed by an ideal transport delay. The drawing supplies a physical interpretation, while the declared parameters define a synthetic teaching model.



One physical interpretation of a first-order-plus-delay model

A stirred thermal inventory followed by ideal transport is one possible physical interpretation of the identified transfer function. The effective conductance includes wall leakage and through-flow heat removal. Its fitted gain and time constant do not separately determine capacity, conductance and heater efficiency.

(a) A synthetic process is $G(s) = Ke^{-\theta s}/(\tau s+1)$ with $K = 1.8$, $\tau = 4$ s and $\theta = 0.6$ s. A 2-unit step starts at $t_0 = 2$ s. Derive the zero-state output and find its final value and the time at which it reaches 63.212 percent of that final value.

Worked reasoning. For $t < 2.6$, output is zero. Thereafter $y(t) = 3.6[1 - e^{-(t-2.6)/4}]$. The final output is 3.6. The fraction $1 - e^{-1}$ occurs one time constant after the delayed onset, at $t = 2 + 0.6 + 4 = 6.6$ s. Measuring the time from zero or from the original step without accounting for delay would overestimate the time constant.

(b) In a deliberately noise-free synthetic step record with amplitude 2, the final output is 3.6, onset occurs 0.6 s after input changes, and the 63.212 percent point occurs 4 s after onset. Recover K, delay and time constant. Explain two obstacles with noisy sampled data.

Worked reasoning. The gain is final change divided by input amplitude: $K = 3.6/2 = 1.8$. Onset identifies $\theta = 0.6$ s, and the post-onset fraction time identifies $\tau = 4$ s. Noise makes a threshold-defined onset ambiguous, while finite sampling quantizes or blurs event timing. A noisy finite record may not reach a clear plateau. Joint model fitting with uncertainty assessment is preferable to treating one threshold reading as exact.

(c) For $C\dot{x} = \eta u - Hx$ followed by pure delay, relate K, τ to C, η, H. Show that $K = 1.8$, $\tau = 4$ do not uniquely identify all three physical parameters.

Worked reasoning. The undelayed transfer is $\eta/(Cs+H)$, so $K = \eta/H$ and $\tau = C/H$. Taking $\eta = 1$ gives $H = 5/9$ and $C = 20/9$ in consistent units. Multiplying η, H, C by any common positive factor leaves both identified ratios unchanged. Consequently a perfect input–output fit does not uniquely recover all physical parameters. The delay is a separate transport idealization; tank geometry is not inferred from the illustration.

2. Fitting a delayed model without leaking validation information

Fit only the training record, using an admissible parameterization. Keep independent excitation and noise for validation, and examine how the numerical delay representation affects optimization.

(a) Write a least-squares objective for fitting K, time constant and delay to sampled step data. Propose a positive-parameter reparameterization and explain its limitation when zero delay is a plausible model.

Worked reasoning. Let $\hat{y}_i = AK[1 - \exp(-\max(t_i - t_0 - \theta, 0)/\tau)]$. Minimize $J = \sum_i (y_i - \hat{y}_i)^2$. Setting $K = e^{\eta_K}$, $\tau = e^{\eta_\tau}$ and $\theta = e^{\eta_\theta}$ enforces strict positivity during unconstrained optimization. It excludes exactly zero delay and assumes positive process gain. If zero delay is plausible, compare a separate zero-delay fit or use bounds allowing $\theta = 0$. The parameterization should express the intended model class rather than silently exclude candidates.

(b) Explain why reporting only training RMSE after fitting a noisy step response gives incomplete evidence. Specify a held-out synthetic input record and noise-generation procedure that avoids reusing the training noise.

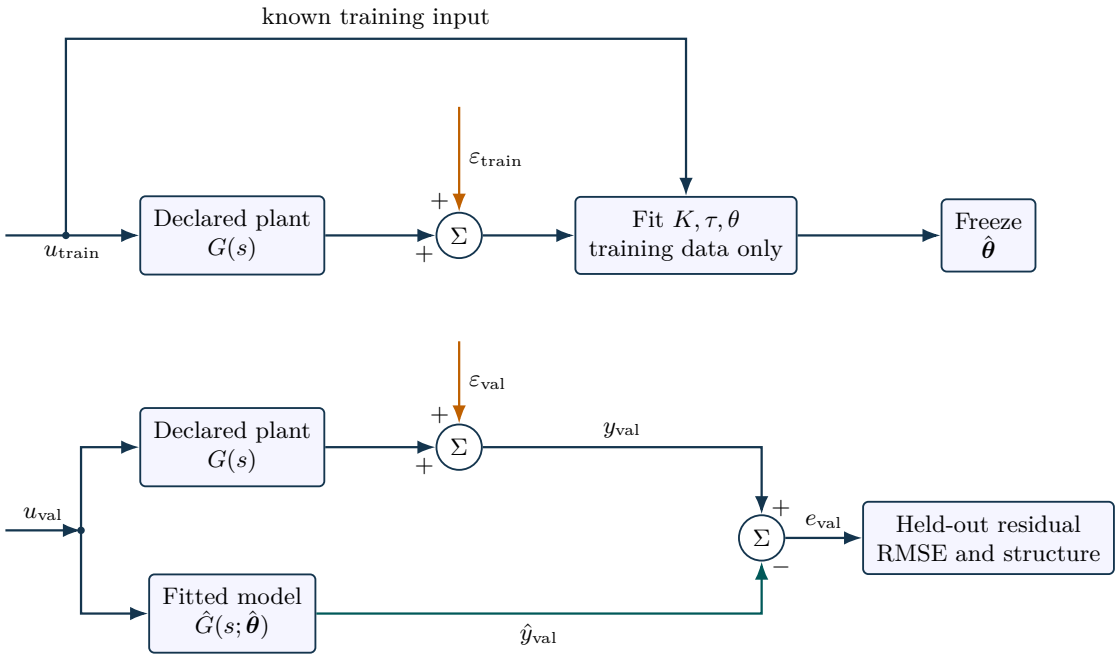
Worked reasoning. The optimizer has already used training residuals to choose parameters, so training error can reward overfitting and reveal little about other excitation patterns. Freeze the fitted parameters, then apply a multistep input with changes at 1, 8, 15 and 22 s and independently generated noise. Evaluate predictions without refitting. Keep the random seed or generator state reproducible while using distinct draws. This tests generalization within a synthetic plant class, not agreement with a real process.

(c) What happens if a gradient-based fit represents delay only by a buffer rounded to an integer number of fixed time steps? Propose a more informative fitting representation and a subsequent implementation check.

Worked reasoning. Rounding makes the predicted output piecewise constant with respect to delay between sample thresholds; its derivative can be zero over intervals and jump at their boundaries. This gives misleading gradient information. Fit the analytic delayed-step response or a justified fractional-delay/interpolation model with explicit bounds. Then test the selected implementation on an independent multistep input, including delay onset and a time-step refinement study. Good optimization of one representation does not validate another automatically.

3. A held-out multistep test with a statistical reference

The validation record uses new input transitions and independent noise. Interpret residual magnitude using a specified stochastic model rather than claiming a perfect model from one small RMSE.



Different input record and independent noise; no refitting on validation data.
All records in this exercise are synthetic.

Training selects parameters; held-out validation freezes them. Separate noise records avoid reusing training residuals as validation evidence.

(a) Use input increments $+1$ at 1 s, -0.6 at 8 s, $+1.1$ at 15 s and -0.8 at 22 s. Write the exact synthetic model output and find its eventual equilibrium.

Worked reasoning. For increments Δu_j at times t_j, superposition gives $y(t) = K\sum_j \Delta u_j[1 - e^{-\max(t-t_j-\theta,0)/\tau}]$. Each term is zero until its delayed onset. The final input level is $1-0.6+1.1-0.8 = 0.7$, so $y_\infty = 1.8(0.7) = 1.26$. Negative increments reduce the input level; they are not negative absolute input levels by themselves.

(b) Validation adds independent zero-mean Gaussian noise of standard deviation 0.015. Explain the expected mean squared residual for a perfect deterministic model. Does a realized RMSE of 0.0145 prove the identified dynamics are exact?

Worked reasoning. For a perfect deterministic prediction, residuals equal noise and $E[r_i^2] = \sigma^2 = 0.000225$. Therefore the expected sample MSE is 0.000225, while finite-sample RMSE fluctuates and its expectation is not exactly σ because of the square root. An RMSE of 0.0145 is compatible with this noise scale but does not prove exact parameters. Bias, time correlation, input-correlated errors and performance on different excitation records need separate inspection.

(c) For a perfect fixed deterministic model and $N = 400$ independent Gaussian validation errors with $\sigma = 0.015$, derive a central 95 percent reference interval for realized RMSE. Explain why it is not a confidence interval proving the fitted dynamics correct.

Worked reasoning. Under exactly these assumptions, $N\,\mathrm{RMSE}^2/\sigma^2 \sim \chi^2_N$. The central interval is $[\sigma\sqrt{\chi^2_{400,0.025}/400}, \sigma\sqrt{\chi^2_{400,0.975}/400}]$. It is approximately $[0.01396, 0.01604]$; the executable check reports precise endpoints. A value 0.0145 is compatible with this noise model, but that is not evidence that every dynamic parameter is correct. Estimated parameters, correlated residuals or model mismatch change the interpretation.

4. Tuning a PI law subject to uncertain steady capability

Treat the tuning rule as a proposal and separately test whether the actuator can maintain the requested operating point under gain uncertainty and a constant disturbance.

(a) Use $k_p = \tau/[K(\lambda + \theta)]$, $k_i = k_p/\tau$ with $\lambda = 2$ s and nominal parameters. Calculate both gains and their changes if the assumed delay rises to 0.9 s.

Worked reasoning. Nominally $k_p = 4/[1.8(2.6)] = 100/117 \approx 0.854701$ and $k_i = 25/117 \approx 0.213675$. At delay 0.9 s, $k_p = 4/[1.8(2.9)] = 200/261 \approx 0.766284$ and $k_i = 50/261 \approx 0.191571$. The larger assumed delay reduces aggressiveness in this tuning rule. These are proposed gains, not a proof of constrained closed-loop stability for every uncertain process.

(b) The process input is constrained to $[0, 2]$, and the reference is one. A constant disturbance of -0.2 is added inside the first-order state equation as $\tau\dot{y} = -y + Ku(t - \theta) - 0.2$. Find the required equilibrium input before and after the disturbance.

Worked reasoning. Before the disturbance, $u_{eq} = 1/K = 5/9 \approx 0.555556$. Afterward $0 = -1 + 1.8u - 0.2$, so $u_{eq} = 1.2/1.8 = 2/3 \approx 0.666667$. Both fit within $[0, 2]$. The disturbance here is in output-equivalent units inside the state balance. A disturbance added to the actuator input would be multiplied by K and require a different compensation calculation.

(c) Suppose the true gain lies in $[1.5, 2.1]$, $u \in [0, 2]$, and the disturbance in the state equation is -0.2. Find the largest reference guaranteed feasible at equilibrium for all allowed gains. Can increasing λ make a reference of 3 feasible?

Worked reasoning. The equilibrium is $y = Ku - 0.2$. The smallest guaranteed maximum output occurs at $K = 1.5$, giving $r_{\max} = 1.5(2) - 0.2 = 2.8$. A reference of 3 requires $u = 3.2/K$, which exceeds 2 at the low-gain endpoint. Changing λ changes the proposed PI gains and response speed, not the physical equilibrium relation or actuator capacity. Feasibility alone does not prove delayed closed-loop stability across the uncertainty set.

5. Implementing a noninteger transport delay

The nominal simulation uses an exactly aligned delay, while a fitted delay need not align with the numerical grid. Preserve this distinction in uncertainty tests and reproduction notes.

(a) Let $h = 0.01$ s and delay 0.6 s, with zero-based indexing. Derive the exact held-input plant update and state which stored control sample is used at index k. Specify the first index at which an initial command u_0 influences the update.

Worked reasoning. There are $N = \theta/h = 60$ delay intervals. With $a = e^{-h/\tau}$, use $y_{k+1} = ay_k + (1 - a)[Ku_{k-N} + d_k]$ for $k \geq N$ and zero delayed input for $k < N$. The first affected update is $k = 60$, mapping y_{60} to y_{61} using u_0. Before that, the initial command is still inside the delay buffer. The sample value at time 0.6 s is unchanged, while its right-hand evolution starts responding.

(b) The fitted parameters are approximately $[1.800025, 3.993491, 0.604289]$. Describe how to evaluate delay and gain uncertainty without claiming physical validation. Include training/validation separation, actuator limits and a numerical convergence check.

Worked reasoning. Freeze the fitted controller, choose an explicitly declared grid or distribution of synthetic gain and delay variations, and run separate closed-loop simulations with the same limits and reference/disturbance schedule. Record stability indicators, tracking error, peak output and time in saturation. Keep the identification record separate from held-out prediction tests. Refine the integration interval and delay representation to check numerical convergence. Report only the tested parameter range and model class; finite simulations do not establish universal robustness or fidelity to real equipment.

(c) For $\theta = qh + \delta$ with integer q and $0 \leq \delta < h$, derive the exact held-input update of the first-order process over one step. Identify both historical input samples used and apply the indexing rule to $\theta = 0.604289$ s, $h = 0.01$ s.

Worked reasoning. Let $a = e^{-h/\tau}$ and $b = e^{-(h-\delta)/\tau}$. During the first δ seconds the delayed input is u_{k-q-1}; during the remainder it is u_{k-q}. Integrating both subintervals gives $y_{k+1} = ay_k + K[(b - a)u_{k-q-1} + (1 - b)u_{k-q}]$. Here $q = 60$ and $\delta = 0.004289$ s. Negative-index inputs use the declared prehistory, typically zero. This exponential weighting is the exact update for the stipulated zero-order hold; simply rounding the delay changes the model.